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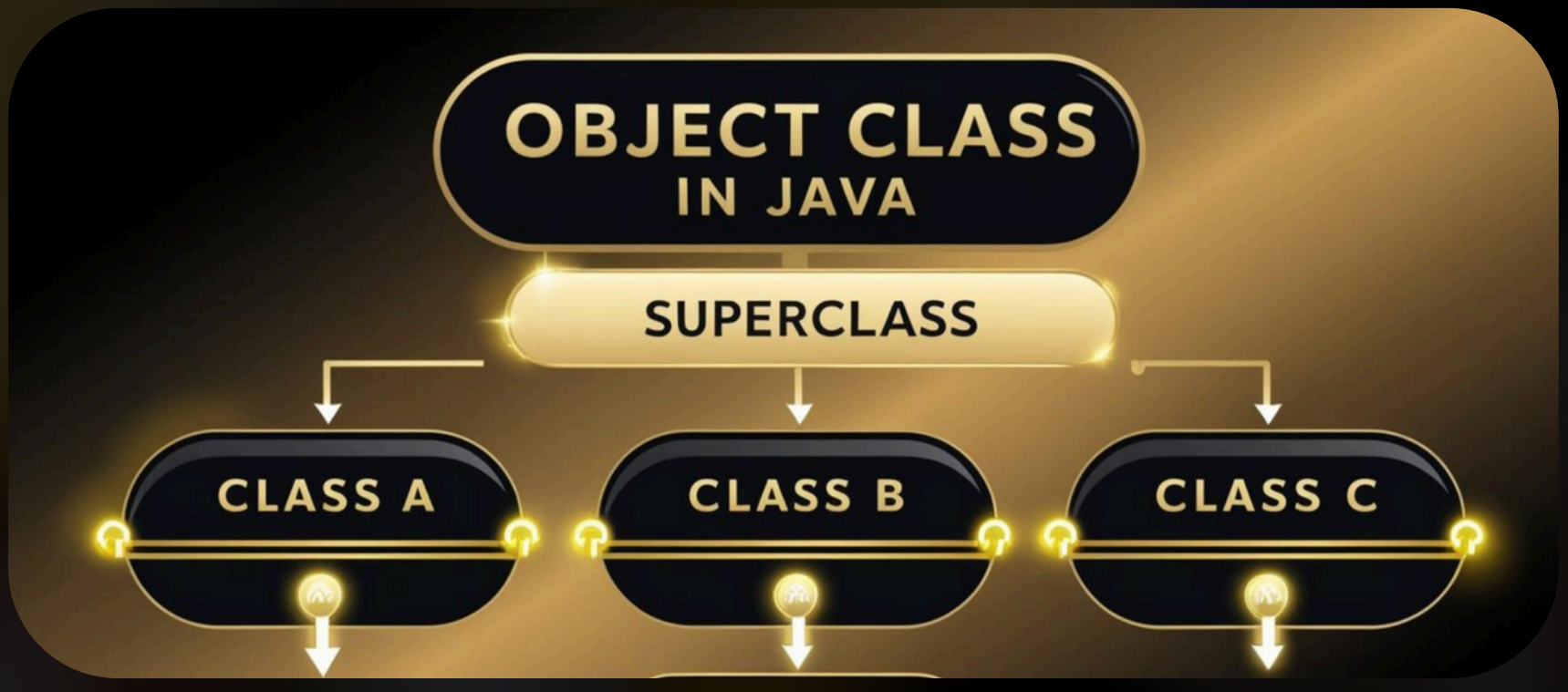
# OBJECT CLASS FAQ




**What is Object class OR Default Super Class?**

- By default, it is a super class for all the classes in java
- All the classes in java are the subclass of object class directly or indirectly
- The Object class defines some common behaviours which is applicable for all the objects
- Example: Cloning mechanism, comparing techniques, Garbage Collections, object can be notified
- Such as
  - Cloning mechanism: object can be compared
  - Object can be cloned
  - Object can be notified etc.
  - Comparing techniques
  - Garbage Collections
- All these features applicable for each and every object in java
- So, if we declare common functionality methods in object class It can be accessed by any java object

**Important methods in Object class**

- Object class has 9 non-static methods
- which can be accessed by any java class

**Object class methods:**

| | | |
|---|---|---|
| toString () | Wait () | Clone () |
| Equals (Object obj) | Wait (long) | Finalize () |
| hashCode () | Wait (long, int) | getClass () |
| | Notify () | |
| | notifyAll () | |

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# OBJECT CLASS METHODS FAQ

**String toString () methods:**

- Defined in object class
- **By default: it** returns the string representation of the object. [ **Format:** Fullyqualifiedclassname@Address]
  we can override the toString () method in the user defined class.
- Use: By overriding this we can get string representation of object

**Why we need to override toString () method:**

- By default: it returns the string representation of the object. [ **Format:** Fullyqualifiedclassname@Address] and these details no use for end user
- Instead of displaying some garbage Address to end user we can override the toString () method in user defined class
- By overriding this we can get string representation of object
- We can display class members and its value to end user

**Equals () Method**

- Defined in Object class
- It is used to check the equality between any two objects
- Based on the hashCode number of objects, it will check the equality between objects
- It is used to compare the values of two objects
- if two objects having same value --- > It returns **TRUE**
- if two objects having different value --- > It returns **FALSE**

**equals () v/s '==' Operator**

- Both are used to check the equality between the two objects
- '== 'operator is used for address comparison
- equals () method is used for content comparison

**hashCode ()**

- returns the hashCode number of the object
- If we invoke hashCode () methods on any object, it returns hashCode number of the object which is based on the hexadecimal address of the object
- We can change the default implementation of this method by overriding it
- But we should provide our own login which generated unique hash code number

**Why we need to override equals () and hashCode () method.**

- "If two objects are equal using Object class equals method then the hashCode method should give the same value for these two objects."
- Hash code of 2 objects same when objects are equal using object class equal method'

- If we **override only equal** method, a. **equals**(b) are true
  - it means the **hashCode** of a and b **must** be same but not happen.
  - Note: **hashCode** () method of Object class always return new **hashCode** for each object.
  - So, when you **need** to use your object in the hashing-based collection, **must override** both **equals** () and **hashCode** ()

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# PURPOSE OF THIS DOCUMENT

- Use this PDF as a quick last-minute revision guide only.

- For more details, please perform R&D and gain a deeper understanding of how memory architecture works.

- Once the interview series is complete, we will cover this topic in detail with code snippets.